Energy Conservation

We know from the Lindblad master equation that energy (i.e. σ_z expectation value) is conserved locally on each spin. The bell states all have $\langle\sigma_{z,i}\rangle_{bell} = 0$. If our initial state has $\langle\sigma_{z,i}\rangle_{initial} = a_i$ what is the maximum possible concurrence? *Left as an exercise* (Sanity check, if $a_1 = a_2 = 0$ then $\mathcal{C}_{max} = 1$, if $|a_1| = |a_2| = 1$ then $\mathcal{C}_{max} = 0$). From this result we can (probably) safely restrict ourselves to initial states with $a_1 = a_2 = 0$

Homodyne detection

In our meeting yesterday we noticed a few things, for an initial product state, $\mathcal{C}_{max} < 1$. However, certain bell states can retain $\mathcal{C} = 1$. Lets try to understand this, I will use the notation from Gabriele's notes. From our photodetection analysis we remember that $\hat{C}_+|\psi_\pm\rangle = 0$, i.e. $|\psi_\pm\rangle$ is an eigenstate of $\hat{C}_+$ with eigenvalue 0. Similarly, $\hat{C}_-|\phi_\pm\rangle = 0$. This means it has no effect in the Stochastic Schrodinger equation. Now we need to understand the effect of the other operator. Let's focus on $|\phi_+\rangle$ but the result should generalise. We want to calculate SSE for $\hat{c} = e^{i\theta}\hat{C}_+$

$$d|\phi_+\rangle = \left\{\left[-1/2\left(\hat{C}_+^2 - 2\cos\theta\langle\hat{C}_+\rangle\hat{C}_+ + \cos^2\theta\langle\hat{C}_+\rangle^2\right)\right]dt + \left(e^{i\theta}\hat{C}_+ - 2\cos\theta\langle\hat{C}_+\rangle\right)dw_t\right\}|\phi_+\rangle$$

This simplifies significantly since $\langle\hat{C}_+\rangle = 0$ so we have

$$d|\phi_+\rangle = -2\gamma dt|\phi_+\rangle - 2\sqrt{\gamma}e^{i\theta}dw_t|\phi_-\rangle$$

From this equation it's a little easier to understand why $\theta = \pi/2$ is optimal. It conserves energy on the trajectory. You can also prove that the updated state is maximally entangled. There is probably an easier way to demonstrate this but this is how I understand it.

Takeaway

This tells me (with a few leaps in logic) that if I start completely in one of the parity subspaces I will stay there. It is completely analogous to qubit dephasing. If $\theta \neq \pi/2$, in the long time limit I end up in either $|01\rangle$ or $|10\rangle$ ($|0\rangle$ or $|1\rangle$ for qubit, $|11\rangle$ or $|00\rangle$ for the other partity subspace). In the qubit case, when $\theta = \pi/2$ and you start in a state of the form $|0\rangle + e^{i\alpha}|1\rangle$ you stay in a state of this form with α undergoing brownian motion. The analogous thing happens for us.

Why 0.5 Concurrence?

In general our operators are $\hat{c}_\pm = e^{i\theta_\pm}\hat{C}_\pm$. It seems that for any initial product state with $a_1 = a_2 = 0$ we get $\mathcal{C} = 0.5$ in the long time limit as long as either $\theta_+ = \pi/2$ or $\theta_- = \pi/2$ but not both, if both we get 0. For any initially maximally entangled state it remains maximally entangled

($\mathcal{C} = 1$) if $\theta_+ = \theta_- = \pi/2$. Understanding this discrepancy will help us understand what is going on. Is it related to this which path idea from Jordan?